

RADIUS vs TACACS+ — AAA Assignment

Task 1. Create a comparison table listing at least 4 key features of RADIUS and TACACS+ side by side, highlighting which protocol supports each feature.

Answer

Feature	RADIUS	TACACS+
Transport Protocol	✓ UDP	✓ TCP
Default Port	UDP 1812 (Authentication) and 1813 (Accounting)	TCP 49
Encryption	 Encrypts the user's password in the authentication exchange; other packet information is not fully encrypted	 Encrypts the TACACS+ packet body, providing broader protection of AAA information
Authentication	✓ Supported	✓ Supported
Authorization	✓ Supported, especially for network access; capabilities depend on the device/server implementation	★ Strong support, including granular command authorization
Accounting	✓ Supported	✓ Supported
Command-by-Command Authorization	⚠ Limited/device-dependent	★ Strong and granular support
Common Use	WiFi, VPN, 802.1X and network access	Router, switch, firewall and network-device administration
AAA Separation	Authentication and authorization are commonly handled together in the RADIUS Access-Accept response	Authentication, authorization and accounting are more clearly separated
Best Suited For	 Network access authentication	 Network administrator/device access

★ **Key Point**

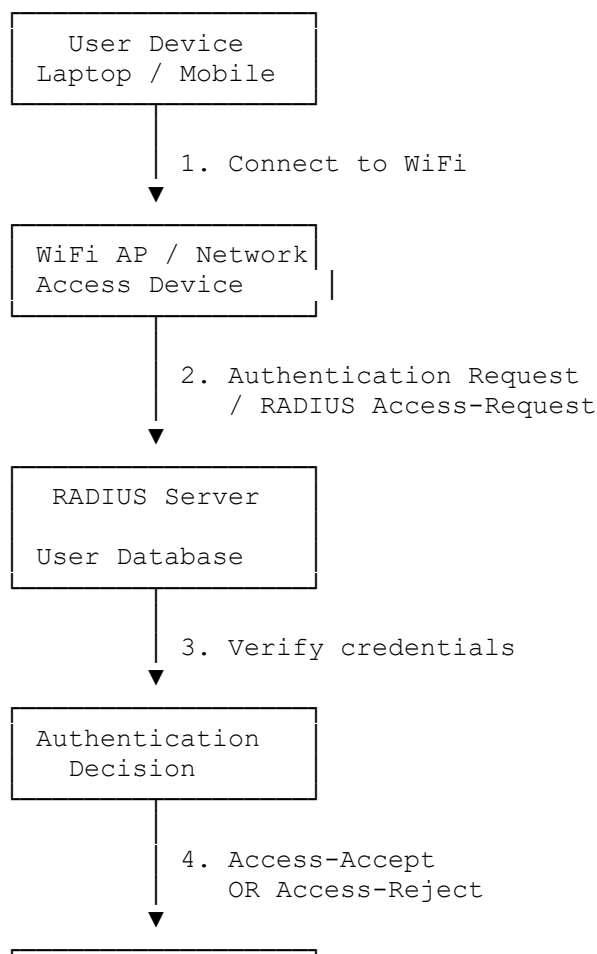
RADIUS → mainly preferred for network access such as WiFi and VPN.

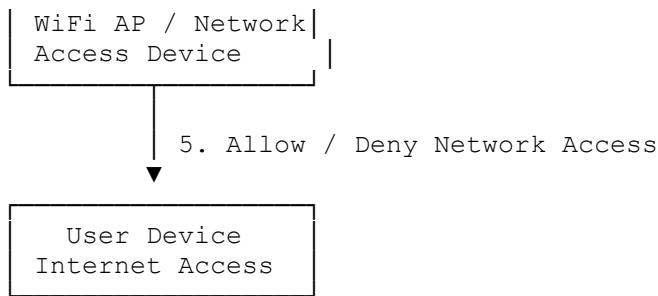
TACACS+ → mainly preferred for administrative access to network devices because of granular command authorization.

Task 2. Draw a step-by-step flow diagram showing the RADIUS authentication process for a user logging in to a public WiFi hotspot, labeling each communication step between the client, network device, and RADIUS server.

Answer

RADIUS Public WiFi Authentication Flow





Step-by-Step Process

Step 1 — WiFi Connection

The user connects a laptop/mobile phone to the college or hotspot WiFi.

Step 2 — Authentication Request

The WiFi access device acts as the **RADIUS client/NAS** and sends an authentication request to the RADIUS server.

Step 3 — Credential Verification

The RADIUS server checks the user's credentials against its configured user database or connected identity system.

Step 4 — Authentication Response

The RADIUS server returns:

-  **Access-Accept** → authentication successful
-  **Access-Reject** → authentication failed
-  **Access-Challenge** → additional authentication information is required

Step 5 — Network Access

If accepted, the access device permits the user to access the network according to the configured policies.

Important Terms

Term	Meaning
Supplicant	User's laptop/mobile or WiFi client
RADIUS Client / NAS	AP, wireless controller or network access device
RADIUS Server	Performs centralized authentication and policy processing
Access-Request	Request sent to the RADIUS server
Access-Accept	Authentication approved
Access-Reject	Authentication denied
Access-Challenge	Additional authentication required

Task 3. List 3 real-world use cases where TACACS+ is preferred over RADIUS, and briefly explain why TACACS+ is better suited for each scenario.

Answer

No.	Real-World Use Case	Why TACACS+ Is Better Suited
1	 Enterprise Router & Switch Administration	TACACS+ provides granular command authorization, allowing administrators to control which commands different users can execute.
2	 Firewall Administration	Security teams can authenticate administrators and apply different authorization levels for configuration and operational commands.
3	 Large Enterprise Network Operations	TACACS+ provides centralized administrator authentication, authorization and accounting, making it easier to control and audit network-device access.

Example

Suppose a company has:

- Junior Network Engineer
- Senior Network Engineer
- Network Administrator

With TACACS+, the organization can implement policies such as:

```
Junior Engineer
  ↓
Show / monitoring commands
  ↓
Limited configuration access

Senior Engineer
  ↓
More configuration commands
  ↓
Broader access

Network Administrator
  ↓
Full administrative commands
```

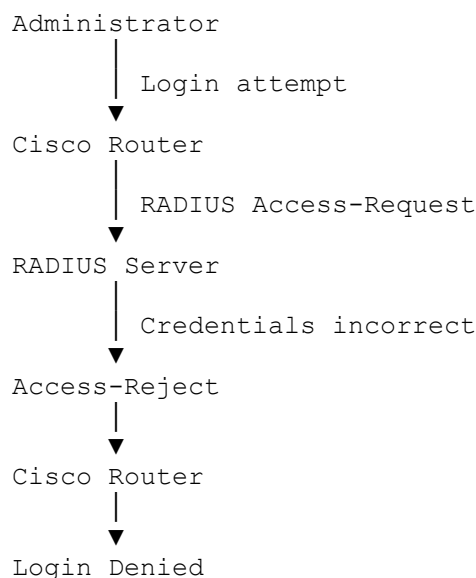
This **command-level authorization** is one of the major reasons TACACS+ is commonly used for network-device administration.

Task 4. Given a scenario where a user tries to log into a network device but fails authentication, describe how RADIUS and TACACS+ would handle the logging and response differently.

Answer

Assume an administrator attempts to log in to a Cisco router with an incorrect username or password.

● RADIUS

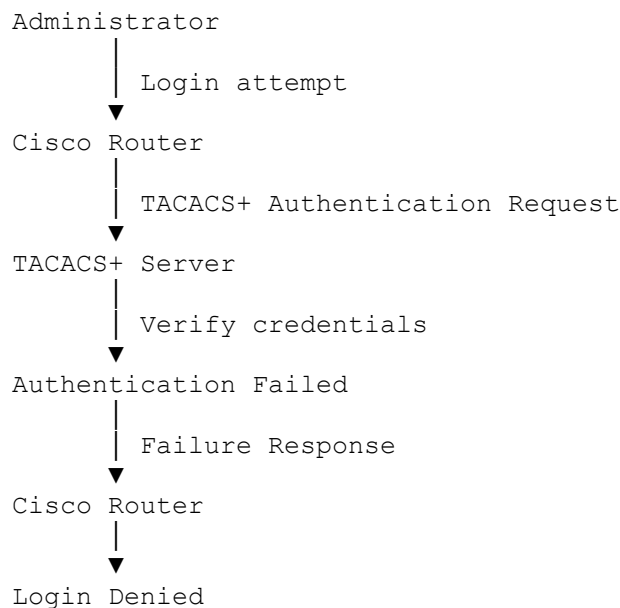


The RADIUS server can record authentication/accounting information such as:

- Username/identity
- Authentication result
- Session information when a session is established
- Login/accounting events

However, **RADIUS is primarily designed around network-access AAA**, such as WiFi and VPN access. Its command authorization capabilities are more limited and implementation-dependent.

TACACS+



TACACS+ can provide detailed administrative AAA, including:

- Authentication result
- Administrator identity
- Authorization decisions
- Command authorization
- Accounting of administrative sessions/activities

Main Difference

Area	RADIUS	TACACS+
Failed authentication response	Access-Reject	Authentication failure/reject
Authentication logging	✓ Yes	✓ Yes
Accounting	✓ Yes	✓ Yes
Detailed command authorization	⚠ Limited/device-dependent	★ Strong
Best fit	Network access	Network-device administration

★ Conclusion

For **WiFi/VPN access**, RADIUS is commonly used because it is designed for centralized network-access authentication.

For **administrative access to routers, switches and firewalls**, TACACS+ is commonly preferred because it provides more granular authorization and accounting capabilities.

Task 5. Use ChatGPT or another AI tool to generate a sample RADIUS configuration snippet for a Cisco router, then review the output and list any changes you would make to adapt it for a college campus WiFi setup.

Answer



AI-Generated Sample Cisco RADIUS Configuration

```
enable
configure terminal

aaa new-model

radius server CAMPUS-RADIUS
  address ipv4 192.168.100.10 auth-port 1812 acct-port 1813
  key CampusRADIUS123

aaa authentication login default group radius local

aaa authorization network default group radius local

aaa accounting network default start-stop group radius

username admin privilege 15 secret LocalBackup123

line vty 0 4
  login authentication default
  transport input ssh

end
write memory
```



Review of the AI-Generated Configuration

The configuration provides a basic RADIUS-based AAA structure, but I would make several changes before using it for a college campus WiFi network.



1 Change the RADIUS Server IP

Current:

```
address ipv4 192.168.100.10
```

Change it to the **actual IP address of the college RADIUS server**.

Example:

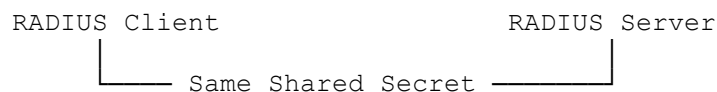
```
address ipv4 10.10.50.10
```

2 Use a Strong Shared Secret

Current:

```
key CampusRADIUS123
```

The shared secret should be replaced with a strong, unique value configured identically on the network device and RADIUS server.



The shared secret should **not be confused with the student's WiFi password**. It is used between the RADIUS client and RADIUS server.

3 Configure the Actual Campus Network Device

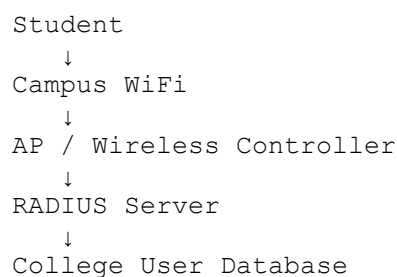
For campus WiFi, the RADIUS client is typically an **AP, wireless LAN controller, or other network-access device**, depending on the architecture.

The Cisco router configuration would therefore be adapted to the actual device performing the RADIUS authentication.

4 Configure Student Authentication

The RADIUS server should connect to the college's identity/user database.

Example:



↓
Accept / Reject

This allows student accounts to be centrally managed.

5 Use Appropriate Authorization Policies

Different users can receive different network permissions.

User Type	Example Authorization
🎓 Student	Internet + approved college resources
👨‍🏫 Faculty	Internet + faculty resources
🔧 IT Staff	Administrative network resources
👤 Guest	Internet-only guest network

6 Configure Accounting

Accounting should be enabled so the college can maintain useful session records, subject to its privacy and retention policies.

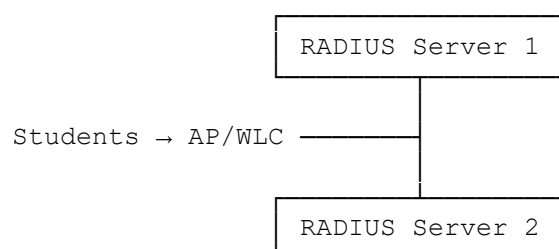
Possible records include:

- Username/account identifier
- Login result
- Session start/end
- Assigned network information
- Session duration
- Accounting status

⚠️ Passwords and other authentication secrets should **never be stored in plaintext logs**.

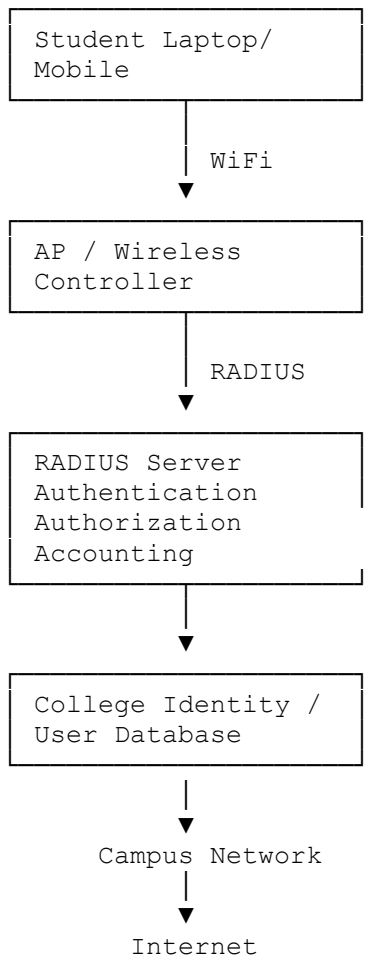
7 Add Redundancy for a Large Campus

For a large college campus, I would avoid relying on only one RADIUS server.



A secondary RADIUS server can provide resilience if the primary server becomes unavailable.

✓ Final Campus WiFi Design



🎯 Overall Conclusion

RADIUS and TACACS+ both provide AAA functionality, but they are commonly used for different purposes.

- **RADIUS** → WiFi, VPN, 802.1X and other network-access authentication.
- **TACACS+** → Administrative access to routers, switches, firewalls and other network devices.
- **RADIUS** uses UDP and commonly uses ports **1812/1813**.
- **TACACS+** uses TCP port **49**.
- **TACACS+** provides particularly strong support for granular command authorization.
- A college campus WiFi deployment would typically use **RADIUS with appropriate redundancy, centralized identity management, authorization policies and accounting.**

